STCS 6701: Probabilistic Machine Learning
Homework 2

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Problem 1

Question 1.1

(a) If $\theta_i \gg \beta_j$ and $\alpha_j > 0$, we have $\alpha_j(\theta_i - \beta_j) \gg 0$, hence the vote is very likely to be "yea".

If θ_i is smaller than β_j, then $\alpha_j(\theta_i - \beta_k) < 0$ hence the vote is more likely to be "nay".

(b) Given two bills with the same β_j, the sign of α_j controls the polarity of the bill. Fix a senator and an ideology difference $\delta := \theta_i - \beta_j$; a positive δ indicates that the senator is more conservative than the bill. A positive value of α_j makes "yea" more likely, while a negative value of α_j makes "nay" more likely; the scenario flips when $\delta < 0$.

A larger magnitude of α_j indicates a more *polarizing* bill: for fixed δ, the mean of z_{ij} is shifted further away from 0 as $|\alpha_j|$ becomes larger, which means that the probability of voting contrary to the expected direction $1(\alpha_j(\theta_i - \beta_j))$ shrinks.

(c) We need only set up our model (local latent variables z_{ij}, ϵ_{ij}) for pairs of indices (i, j) for which y_{ij} is *not* missing (or held-out). Call the collection of all such indices $\mathcal{I}$; the joint likelihood for our model only contains terms from $\mathcal{I}$, and looks like

$$p(\boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) \prod_{(i,j)\in\mathcal{I}} p(y_{ij} \mid \theta_i, \alpha_j, \beta_j).$$

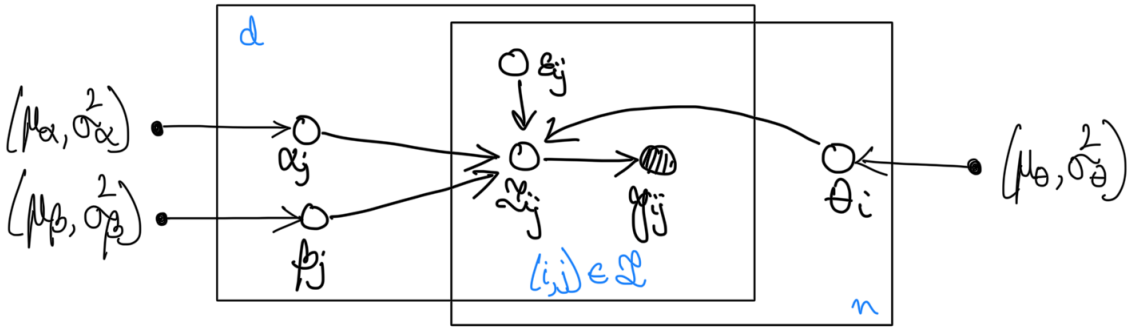
In particular,

$$p(y_{ij} = 1 \mid \theta_i, \alpha_j, \beta_j) = \mathbb{P}(z_{ij} > 0 \mid \theta_i, \alpha_j, \beta_j) = \Phi(\alpha_j(\theta_i - \beta_j)).$$

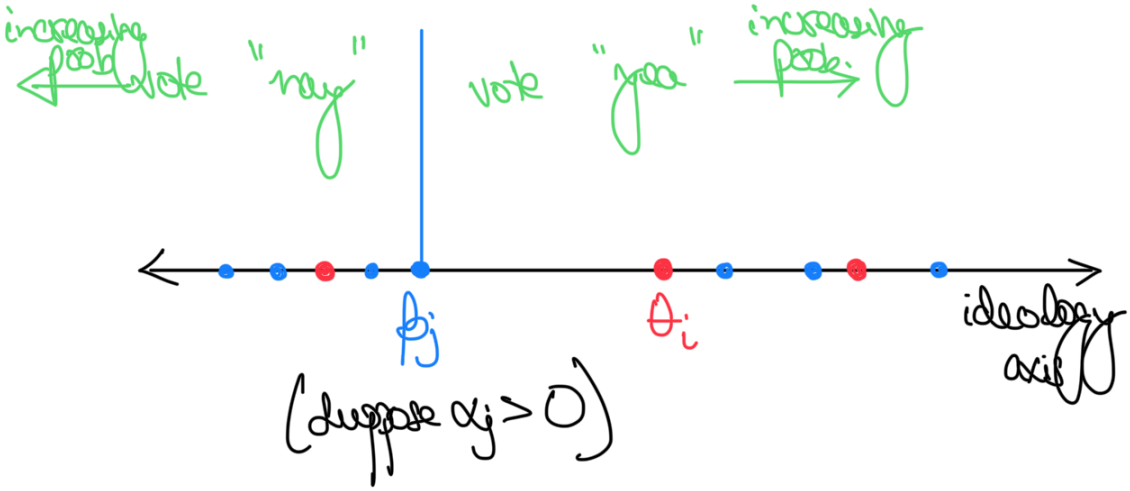
For $(i, j) \notin \mathcal{I}$, one can of course evaluate the posterior predictive likelihood

$$p(y_{ij} \mid \boldsymbol{y}_{in}) = \int p(y_{ij} \mid \theta_i, \alpha_j, \beta_j)\, p(\theta_i, \alpha_j, \beta_j \mid \boldsymbol{y}_{in})\, d\theta_i\, d\alpha_j\, d\beta_j.$$

(d) The model can be represented using plate notation as follows.



(e) Senator ideological positions θ_i have been marked in red, bill positions β_j in blue on the same ideological axis.



For the particular bill j, suppose that $\alpha_j > 0$ for simplicity. This determines a threshold: a senator i with position $\theta_i > \beta_j$ is more likely than not to vote "yea", with the likelihood increasing the further away θ_i is from the threshold. Conversely, senators to the left of the threshold are more likely to vote "nay".

(f) The held-out data (indexed by $\mathcal{I}_o$) can be used to compute a goodness of fit score,

such as an average predictive log-likelihood

$$\sum_{(i,j)\in\mathcal{I}_o} \int \log p(y_{ij} \mid \theta_i, \alpha_j, \beta_j)\, p(\theta_i, \alpha_j, \beta_j \mid \boldsymbol{y}_{in})\, d\theta_i\, d\alpha_j\, d\beta_j.$$

The (approximately inferred plug-in) posterior depends on the prior variances; maximizing this score over choices of these prior parameters is an Empirical Bayes approach to setting them.

Question 1.2

1. Denoting the shifted variable with accents, note that the model likelihood

$$\begin{aligned}\mathcal{L}(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) &= \prod_{(i,j)\in\mathcal{I}} p(y_{ij} \mid \theta_i, \alpha_j, \beta_j) \\ &= \prod_{(i,j)\in\mathcal{I}} \Phi(\alpha_j(\theta_i - \beta_j))^{y_{ij}} (1 - \Phi(\alpha_j(\theta_i - \beta_j)))^{1-y_{ij}}\end{aligned}$$

depends on $\boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}$ only through the quantities $\mu_{ij} := \alpha_j(\theta_i - \beta_j)$. Since

$$\alpha_j(\theta'_i - \beta'_j) = \alpha_j((\theta_i + c) - (\beta_j + c)) = \alpha_j(\theta_i - \beta_j),$$

the likelihood $\mathcal{L}$ remains unchanged after making the indicated replacements.

This means that raw values on the ideological axis have no absolute interpretation; we may as well fix the ideological value of one item (a senator or a bill) and measure the ideology of all other items (senators and bills) relative to them. The model only deals with differences in ideology $\theta_i - \alpha_j$.

2. Using the same setup,

$$\alpha'_j(\theta'_i - \beta'_j) = \left(\frac{\alpha_j}{k}\right)(k\theta_i - k\beta_j) = \alpha_j(\theta_i - \beta_j)$$

shows that $\mathcal{L}$ remains unchanged after making the indicated rescaling.

Again, this means that the absolute scale of the ideological axis is arbitrary; we may as well fix the absolute value of the polarity α_j of one bill (say to 1) and measure the polarities of all other bills relative to it.

3. The symmetries in the parameters indicate that they can only be interpreted in a relative sense (differences in ideologies, ratios of discriminations); the raw values of these parameters are somewhat arbitrary.

Combining the above symmetries shows that the model likelihood is invariant under the transformation

$$\alpha_j \mapsto \alpha_j/k, \qquad \beta_j \mapsto k\beta_j + c, \qquad \theta_i \mapsto k\theta_i + c.$$

The suggested choice of prior will of course break this arbitrariness, favoring shrinkage towards zero in the posterior.

Question 1.3

(a) Note that $y_{ij} \mid z_{ij}$ is completely determined by $1(z_{ij} > 0)$, with

$$p(y_{ij} \mid z_{ij}) = 1(z_{ij} > 0)^{y_{ij}}\, 1(z_{ij} \le 0)^{1-y_{ij}}.$$

Remark: Here we abuse notation with $0^0 := 1$. This uses the interpretation $y_{ij} \mid z_{ij} \sim$ Bernoulli$(1(z_{ij} > 0))$. More precisely, $y_{ij} \mid z_{ij} \sim \delta_{1(z_{ij}>0)}$ since $y_{ij} = 1(z_{ij} > 0)$.

Thus, the joint density is described by

$$\begin{aligned} p(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) &= \prod_{i=1}^{n} p(\theta_i) \prod_{j=1}^{d} p(\alpha_j)\, p(\beta_j) \prod_{(i,j)\in\mathcal{I}} p(z_{ij} \mid \theta_i, \alpha_j, \beta_j)\, p(y_{ij} \mid z_{ij}) \\ &= \prod_{i=1}^{n} (2\pi\sigma_\theta^2)^{-1/2} e^{-(\theta_i-\mu_\theta)^2/2\sigma_\theta^2} \\ &\quad \cdot \prod_{j=1}^{d} (2\pi\sigma_\alpha^2)^{-1/2} e^{-(\alpha_j-\mu_\alpha)^2)/2\sigma_\alpha^2} \cdot (2\pi\sigma_\beta^2)^{-1/2} e^{-(\beta_j-\mu_\beta)^2)/2\sigma_\beta^2} \\ &\quad \cdot \prod_{(i,j)\in\mathcal{I}} (2\pi)^{-1/2} e^{-(z_{ij}-\alpha_j(\theta_i-\beta_j))^2/2} \cdot 1(z_{ij} > 0)^{y_{ij}}\, 1(z_{ij} \le 0)^{1-y_{ij}}. \end{aligned} \tag{$\star$}$$

This is zero when any $y_{ij} \neq 1(z_{ij} > 0)$, and is a product of Gaussian densities otherwise.

(b) Denote $\mu_{ij} = \alpha_j(\theta_i - \beta_j)$. Fixing all terms apart from z_{ij} in the joint $(\star)$ gives

$$p(z_{ij} \mid -) \propto e^{-(z_{ij}-\mu_{ij})^2/2} 1(z_{ij} > 0)^{y_{ij}} 1(z_{ij} \le 0)^{1-y_{ij}}.$$

In particular, this shows

$$p(z_{ij} \mid y_{ij}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) \propto e^{-(z_{ij}-\mu_{ij})^2/2} \cdot \begin{cases} 1(z_{ij} > 0), & \text{if } y_{ij} = 1, \\ 1(z_{ij} \le 0), & \text{if } y_{ij} = 0, \end{cases}$$

which means that

$$z_{ij} \mid y_{ij}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta} \sim \begin{cases} N(\mu_{ij}, 1) \text{ truncated to } (0, \infty), & \text{if } y_{ij} = 1, \\ N(\mu_{ij}, 1) \text{ truncated to } (-\infty, 0], & \text{if } y_{ij} = 0. \end{cases}$$

This assumes that $(i, j) \in \mathcal{I}$, i.e. that y_{ij} was observed. If not, y_{ij} is not part of our model (does not appear in the joint distribution), so there is no question of a corresponding z_{ij} appearing anywhere.

(c) Fixing all terms apart from θ_i in the joint density of $(\boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ (obtained by simply dropping $\boldsymbol{y}$ from the generative process described by $(\star)$) gives

$$p(\theta_i \mid \boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}) \propto e^{-(\theta_i-\mu_\theta)^2/2\sigma_\theta^2} \prod_{j:(i,j)\in\mathcal{I}} e^{-(z_{ij}-\alpha_j(\theta_i-\beta_j))^2/2}.$$

The logarithm of this density is a negative definite quadratic in θ_i, hence this must describe a Gaussian $N(\mu_{\theta_i}, \sigma^2_{\theta_i})$. In particular, the logarithm of the above expression is

$$\begin{aligned} h(\theta_i) &:= -\frac{1}{2\sigma^2_\theta}(\theta_i - \mu_\theta)^2 - \frac{1}{2}\sum_{j:(i,j)\in\mathcal{I}} (z_{ij} - \alpha_j(\theta_i - \beta_j))^2 \\ &=: -\frac{1}{2\sigma^2_{\theta_i}}(\theta_i - \mu_{\theta_i})^2 + C(\boldsymbol{z}, \boldsymbol{\alpha}, \boldsymbol{\beta}). \end{aligned}$$

Then,

$$-h'(\theta_i) = \frac{1}{\sigma^2_\theta}(\theta_i - \mu_\theta) - \sum_{j:(i,j)\in\mathcal{I}} \alpha_j(z_{ij} - \alpha_j(\theta_i - \beta_j)) = \frac{1}{\sigma^2_{\theta_i}}(\theta_i - \mu_{\theta_i}),$$

so comparing coefficients gives the parameters

$$\sigma^{-2}_{\theta_i} = \sigma^{-2}_\theta + \sum_{j:(i,j)\in\mathcal{I}} \alpha_j^2, \qquad \mu_{\theta_i} = \sigma^2_{\theta_i}\left(\sigma^{-2}_\theta \mu_\theta + \sum_{j:(i,j)\in\mathcal{I}} \alpha_j(z_{ij} + \alpha_j\beta_j)\right).$$

(d) Repeating the same arguments from above on β_j, we have

$$p(\beta_j \mid \boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}) \propto e^{-(\beta_j - \mu_\beta)^2/2\sigma^2_\beta} \prod_{i:(i,j)\in\mathcal{I}} e^{-(z_{ij} - \alpha_j(\theta_i - \beta_j))^2/2}.$$

Again, the log-density is a negative definite quadratic in β_j, hence this describes a Gaussian $N(\mu_{\beta_j}, \sigma^2_{\beta_j})$. Calling the logarithm of the above expression $h(\beta_j)$, we match

$$-h'(\beta_j) = \frac{1}{\sigma^2_\beta}(\beta_j - \mu_\beta) + \sum_{i:(i,j)\in\mathcal{I}} \alpha_j(z_{ij} - \alpha_j(\theta_i - \beta_j)) =: \frac{1}{\sigma^2_{\beta_j}}(\beta_j - \mu_{\beta_j}),$$

so comparing coefficients gives

$$\sigma^{-2}_{\beta_j} = \sigma^{-2}_\beta + \sum_{i:(i,j)\in\mathcal{I}} \alpha_j^2, \qquad \mu_{\beta_j} = \sigma^2_{\beta_j}\left(\sigma^{-2}_\beta \mu_\beta - \sum_{i:(i,j)\in\mathcal{I}} \alpha_j(z_{ij} - \alpha_j\theta_i)\right).$$

(e) Repeating the same arguments from above on α_j, we have

$$p(\alpha_j \mid \boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\beta}) \propto e^{-(\alpha_j - \mu_\alpha)^2/2\sigma^2_\alpha} \prod_{i:(i,j)\in\mathcal{I}} e^{-(z_{ij} - \alpha_j(\theta_i - \beta_j))^2/2}.$$

Again, the log-density is a negative definite quadratic in α_j, hence describes a Gaussian $N(\mu_{\alpha_j}, \sigma^2_{\alpha_j})$. Calling the logarithm of the above expression $h(\alpha_j)$, we match

$$-h'(\alpha_j) = \frac{1}{\sigma^2_\alpha}(\alpha_j - \mu_\alpha) - \sum_{i:(i,j)\in\mathcal{I}} (\theta_i - \beta_j)(z_{ij} - \alpha_j(\theta_i - \beta_j)) =: \frac{1}{\sigma^2_{\alpha_j}}(\alpha_j - \mu_{\alpha_j}),$$

so comparing coefficients gives

$$\sigma^{-2}_{\alpha_j} = \sigma^{-2}_\alpha + \sum_{i:(i,j)\in\mathcal{I}} (\theta_i - \beta_j)^2, \qquad \mu_{\alpha_j} = \sigma^2_{\alpha_j}\left(\sigma^{-2}_\alpha \mu_\alpha + \sum_{i:(i,j)\in\mathcal{I}} (\theta_i - \beta_j) z_{ij}\right).$$

(f) We have already computed

$$y_{ij} \mid \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta} \sim \text{Bernoulli}(\Phi(\mu_{ij})),$$

and the 'collapsed' likelihood $p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta})$ in Q1.2.1. Restating this here,

$$p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{(i,j)\in\mathcal{I}} \Phi(\mu_{ij})^{y_{ij}} (1 - \Phi(\mu_{ij}))^{1-y_{ij}},$$

hence

$$\begin{aligned} p(\boldsymbol{y}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{i=1}^{n} (2\pi\sigma_\theta^2)^{-1/2} e^{-(\theta_i - \mu_\theta)^2/2\sigma_\theta^2} \\ \cdot \prod_{j=1}^{d} (2\pi\sigma_\alpha^2)^{-1/2} e^{-(\alpha_j - \mu_\alpha)^2)/2\sigma_\alpha^2} \cdot (2\pi\sigma_\beta^2)^{-1/2} e^{-(\beta_j - \mu_\beta)^2)/2\sigma_\beta^2} \\ \cdot \prod_{(i,j)\in\mathcal{I}} \Phi(\mu_{ij})^{y_{ij}} (1 - \Phi(\mu_{ij}))^{1-y_{ij}}. \end{aligned}$$

(g) Our variational family is described by

$$\begin{aligned} q(\theta_i) &\sim N(\bar{\mu}_{\theta_i}, \bar{\sigma}^2_{\theta_i}), && 1 \le i \le n, \\ q(\alpha_j) &\sim N(\bar{\mu}_{\alpha_j}, \bar{\sigma}^2_{\alpha_j}), && 1 \le j \le d, \\ q(\beta_j) &\sim N(\bar{\mu}_{\beta_j}, \bar{\sigma}^2_{\beta_j}), && 1 \le j \le d, \\ q(z_{ij}) &\sim N(\bar{\mu}_{ij}, 1) \text{ truncated to } \begin{cases} (0,\infty), & \text{if } y_{ij} = 1, \\ (-\infty, 0], & \text{if } y_{ij} = 0, \end{cases} && (i,j) \in \mathcal{I}. \end{aligned}$$

Each update will require the q-expectation of the log densities (discarding 'constants' not involving the parameter being updated) of the full-conditionals which we have computed in parts (b)-(e). Thus, the updates for

- $q(z_{ij})$ require $\mathbb{E}_q[\mu_{ij}] = \mathbb{E}_q[\alpha_j](\mathbb{E}_q[\theta_i] - \mathbb{E}_q[\beta_j])$,
- $q(\theta_i)$ require $\mathbb{E}_q[z_{ij}]$, $\mathbb{E}_q[\beta_j]$, $\mathbb{E}_q[\alpha_j]$, $\mathbb{E}_q[\alpha_j^2]$,
- $q(\alpha_j)$ require $\mathbb{E}_q[z_{ij}]$, $\mathbb{E}_q[\theta_i]$, $\mathbb{E}_q[\theta_i^2]$, $\mathbb{E}_q[\beta_j]$, $\mathbb{E}_q[\beta_j^2]$,
- $q(\beta_j)$ require $\mathbb{E}_q[z_{ij}]$, $\mathbb{E}_q[\theta_i]$, $\mathbb{E}_q[\alpha_j]$, $\mathbb{E}_q[\alpha_j^2]$.

(h) (i) Reusing our work from part (b), the log-density of the full-conditional of z_{ij} is quadratic (restricted to $\mathbb{R}_\pm$ based on y_{ij}) in z_{ij} with coefficients $-\frac{1}{2}$ and μ_{ij}. Taking expectations, we obtain $\bar{\mu}_{ij} = \mathbb{E}_q[\mu_{ij}] = \bar{\mu}_{\alpha_j}(\bar{\mu}_{\theta_i} - \bar{\mu}_{\beta_j})$.

With this,

$$\mathbb{E}_q[z_{ij}] = \bar{\mu}_{ij} + \phi(\bar{\mu}_{ij}) \cdot \begin{cases} 1/\Phi(\bar{\mu}_{ij}), & \text{if } y_{ij} = 1, \\ -1/(1 - \Phi(\bar{\mu}_{ij})), & \text{if } y_{ij} = 0. \end{cases}$$

(ii) Reusing our work from part (c), the log-density of the full-conditional of θ_i is quadratic in θ_i; taking expectations and comparing coefficients on the θ_i-derivative of the log density again (these operations commute!), we have

$$\bar{\sigma}_{\theta_i}^{-2} = \sigma_\theta^{-2} + \sum_{j:(i,j)\in\mathcal{I}} \mathbb{E}_q[\alpha_j^2], \quad \bar{\mu}_{\theta_i} = \bar{\sigma}_{\theta_i}^2 \left(\sigma_\theta^{-2}\mu_\theta + \sum_{j:(i,j)\in\mathcal{I}} (\mathbb{E}_q[\alpha_j]\mathbb{E}_q[z_{ij}] + \mathbb{E}_q[\alpha_j^2]\mathbb{E}_q[\beta_j]) \right).$$

Applying precisely the same arguments on the full-conditional of β_j gives the updates

$$\bar{\sigma}_{\beta_j}^{-2} = \sigma_\beta^{-2} + \sum_{i:(i,j)\in\mathcal{I}} \mathbb{E}_q[\alpha_j^2], \quad \bar{\mu}_{\beta_j} = \bar{\sigma}_{\beta_j}^2 \left(\sigma_\beta^{-2}\mu_\beta - \sum_{i:(i,j)\in\mathcal{I}} (\mathbb{E}_q[\alpha_j]\mathbb{E}_q[z_{ij}] - \mathbb{E}_q[\alpha_j^2]\mathbb{E}_q[\theta_i]) \right).$$

(iii) Repeating the same arguments on the full-conditional density of α_j gives the updates

$$\bar{\sigma}_{\alpha_j}^{-2} = \sigma_\alpha^{-2} + \sum_{i:(i,j)\in\mathcal{I}} \mathbb{E}_q[(\theta_i - \beta_j)^2], \quad \bar{\mu}_{\alpha_j} = \bar{\sigma}_{\alpha_j}^2 \left(\sigma_\alpha^{-2}\mu_\alpha + \sum_{i:(i,j)\in\mathcal{I}} (\mathbb{E}_q[\theta_i] - \mathbb{E}_q[\beta_j])\mathbb{E}_q[z_{ij}] \right).$$

Note that

$$\mathbb{E}_q[(\theta_i - \beta_j)^2] = \mathbb{E}_q[\theta_i^2] + \mathbb{E}_q[\beta_j^2] - 2\mathbb{E}_q[\theta_i]\mathbb{E}_q[\beta_j].$$

Probit Ideal Point Modeling for Voting Patterns

Abstract

We apply a probit ideal point model on a dataset of roll-call votes from the 113th U.S. Senate to model voting patterns, and reveal latent ideological structures inherent to both senators and bills. Approximate inference is performed using a Coordinate Ascent Variational Inference (CAVI) algorithm. The inferred ideologies are easily interpretable as degrees of political left/right inclination.

Introduction

Voting patterns in the U.S. Senate are determined by political ideology. An ideal point model quantifies this latent ideology for both senators and bills on a one-dimensional scale, with the difference between a senator's ideological position and the bill's location monotonically affecting the probability of voting "yea". We analyze the 113th U.S. Senate roll-call votes using a probit ideal point model, and observe that the inferred ideological axis is consistent with senators' party affiliation and left/right wing political inclination. This supports the conclusion that voting patterns are largely driven by the latter factors. The results have been presented in Figure 1 for senators, Figure 2 for bills.

Model

Fix $n, d \in \mathbb{N}$, and let $\mathcal{I} \subseteq \{1, \dots, n\} \times \{1, \dots, d\}$ be the collection of pairs (i, j) for which the i-th senator has voted on the j-th bill. Setting prior parameters $\mu_\theta, \mu_\alpha, \mu_\beta \in \mathbb{R}$, $\sigma_\theta^2, \sigma_\alpha^2, \sigma_\beta^2 \in \mathbb{R}^+$, and denoting $\mu_{ij} := \alpha_j(\theta_i - \beta_j)$, consider the following generative model.

$$\begin{aligned}
\theta_i &\sim \text{Normal}(\mu_\theta, \sigma_\theta^2), && 1 \le i \le n, \\
\alpha_j &\sim \text{Normal}(\mu_\alpha, \sigma_\alpha^2), && 1 \le j \le d, \\
\beta_j &\sim \text{Normal}(\mu_\beta, \sigma_\beta^2), && 1 \le j \le d, \\
z_{ij} \mid \theta_i, \alpha_j, \beta_j &\sim \text{Normal}(\mu_{ij}, 1), && (i, j) \in \mathcal{I} \\
y_{ij} \mid z_{ij} &= 1(z_{ij} > 0), && (i, j) \in \mathcal{I}.
\end{aligned}$$

Abusing notation with $0^0 := 1$, the joint density of this model is

$$\begin{aligned}
p(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{i=1}^{n} &(2\pi\sigma_\theta^2)^{-1/2} e^{-(\theta_i - \mu_\theta)^2/2\sigma_\theta^2} \\
&\cdot \prod_{j=1}^{d} (2\pi\sigma_\alpha^2)^{-1/2} e^{-(\alpha_j - \mu_\alpha)^2)/2\sigma_\alpha^2} \cdot (2\pi\sigma_\beta^2)^{-1/2} e^{-(\beta_j - \mu_\beta)^2)/2\sigma_\beta^2} \\
&\cdot \prod_{(i,j) \in \mathcal{I}} (2\pi)^{-1/2} e^{-(z_{ij} - \alpha_j(\theta_i - \beta_j))^2/2} \cdot 1(z_{ij} > 0)^{y_{ij}} \, 1(z_{ij} \le 0)^{1 - y_{ij}}.
\end{aligned}$$

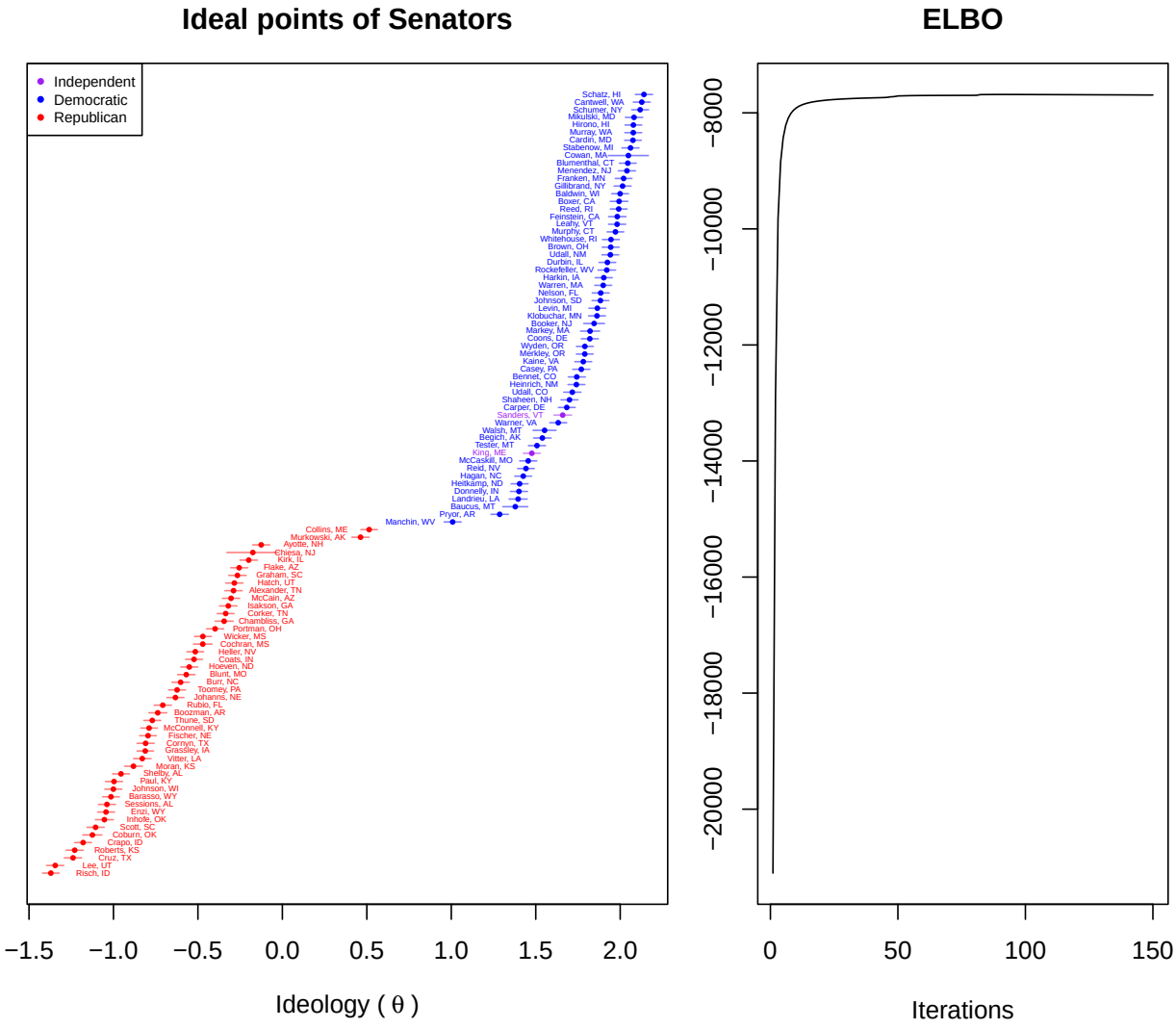


Figure 1: *Left:* Approximate posterior ideologies of senators, depicted as $\bar{\mu}_{\theta_i} \pm 2\bar{\sigma}_{\theta_i}$. This is consistent with party affiliation, denoted by color. The large error bars on Cowan, Chiesa correspond to the unusually small number of votes they cast (154, 70). *Right:* Evolution of the ELBO of the mean-field variational posterior q_ν over iterations.

This is zero when any $y_{ij} \neq 1(z_{ij} > 0)$, and is a product of Gaussian densities otherwise.

Each senator's ideological position θ_i interacts with a bill's ideological threshold β_j and polarizing power α_j to give the likelihood $\Phi(\alpha_j(\theta_i - \beta_j))$ of a vote "yea" in the pair (i, j).

Inference

We use Mean-Field Variational Inference with a Coordinate Ascent Variational Inference (CAVI) algorithm to approximate the posterior distribution of $\boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta} \mid \boldsymbol{y}$. We use a

variational family

$$q_\nu(\boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) = \prod_{i=1}^{n} q(\theta_i; \bar{\mu}_{\theta_i}, \bar{\sigma}^2_{\theta_i}) \prod_{j=1}^{d} q(\alpha_j; \bar{\mu}_{\alpha_j}, \bar{\sigma}^2_{\alpha_j})\, q(\beta_j; \bar{\mu}_{\beta_j}, \bar{\sigma}^2_{\beta_j}) \prod_{(i,j)\in\mathcal{I}} q(z_{ij}; \bar{\mu}_{ij})$$

parametrized by

$$\nu := (\{\bar{\mu}_{\theta_i}\}_{i=1}^n, \{\bar{\sigma}^2_{\theta_i}\}_{i=1}^n, \{\bar{\mu}_{\alpha_j}\}_{j=1}^d, \{\bar{\sigma}^2_{\alpha_j}\}_{j=1}^d, \{\bar{\mu}_{\beta_j}\}_{j=1}^d, \{\bar{\sigma}^2_{\beta_j}\}_{j=1}^d, \{\bar{\mu}_{ij}\}_{(i,j)\in\mathcal{I}}),$$

where

$$\begin{aligned}
q(\theta_i) &\sim N(\bar{\mu}_{\theta_i}, \bar{\sigma}^2_{\theta_i}), && 1 \le i \le n,\\
q(\alpha_j) &\sim N(\bar{\mu}_{\alpha_j}, \bar{\sigma}^2_{\alpha_j}), && 1 \le j \le d,\\
q(\beta_j) &\sim N(\bar{\mu}_{\beta_j}, \bar{\sigma}^2_{\beta_j}), && 1 \le j \le d,\\
q(z_{ij}) &\sim N(\bar{\mu}_{ij}, 1) \text{ truncated to } \begin{cases} (0,\infty), & \text{if } y_{ij} = 1,\\ (-\infty, 0], & \text{if } y_{ij} = 0, \end{cases} && (i,j) \in \mathcal{I}.
\end{aligned}$$

This implies the coordinate-wise updates (to be applied in succession)

$$\begin{aligned}
\bar{\mu}_{ij} &= \bar{\mu}_{\alpha_j}(\bar{\mu}_{\theta_i} - \bar{\mu}_{\beta_j}),\\
\bar{\sigma}^{-2}_{\theta_i} &= \sigma_\theta^{-2} + \sum_{j:(i,j)\in\mathcal{I}} (\bar{\mu}^2_{\alpha_j} + \bar{\sigma}^2_{\alpha_j}),\\
\bar{\mu}_{\theta_i} &= \bar{\sigma}^2_{\theta_i} \left(\sigma_\theta^{-2}\mu_\theta + \sum_{j:(i,j)\in\mathcal{I}} (\bar{\mu}_{\alpha_j}\mathbb{E}_q[z_{ij}] + (\bar{\mu}^2_{\alpha_j} + \bar{\sigma}^2_{\alpha_j})\bar{\mu}_{\beta_j}) \right)\\
\bar{\sigma}^{-2}_{\beta_j} &= \sigma_\beta^{-2} + \sum_{i:(i,j)\in\mathcal{I}} (\bar{\mu}^2_{\alpha_j} + \bar{\sigma}^2_{\alpha_j}),\\
\bar{\mu}_{\beta_j} &= \bar{\sigma}^2_{\beta_j} \left(\sigma_\beta^{-2}\mu_\beta - \sum_{i:(i,j)\in\mathcal{I}} (\bar{\mu}_{\alpha_j}\mathbb{E}_q[z_{ij}] - (\bar{\mu}^2_{\alpha_j} + \bar{\sigma}^2_{\alpha_j})\bar{\mu}_{\theta_i}) \right)\\
\bar{\sigma}^{-2}_{\alpha_j} &= \sigma_\alpha^{-2} + \sum_{i:(i,j)\in\mathcal{I}} (\bar{\mu}^2_{\theta_i} + \bar{\sigma}^2_{\theta_i} + \bar{\mu}^2_{\beta_j} + \bar{\sigma}^2_{\beta_j} - 2\bar{\mu}_{\theta_i}\bar{\mu}_{\beta_j}),\\
\bar{\mu}_{\alpha_j} &= \bar{\sigma}^2_{\alpha_j} \left(\sigma_\alpha^{-2}\mu_\alpha + \sum_{i:(i,j)\in\mathcal{I}} (\bar{\mu}_{\theta_i} - \bar{\mu}_{\beta_j})\mathbb{E}_q[z_{ij}] \right).
\end{aligned}$$

Here,

$$\mathbb{E}_q[z_{ij}] = \bar{\mu}_{ij} + \phi(\bar{\mu}_{ij}) \cdot \begin{cases} 1/\Phi(\bar{\mu}_{ij}), & \text{if } y_{ij} = 1,\\ -1/(1 - \Phi(\bar{\mu}_{ij})), & \text{if } y_{ij} = 0. \end{cases}$$

The Evidence Lower Bound (ELBO) can be directly computed as

$$\begin{aligned}
\text{ELBO}(q_\nu) &:= \mathbb{E}_q[\log p(\boldsymbol{y}, \boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta}) - \log q(\boldsymbol{z}, \boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta})] \\
&\overset{c}{=} \frac{1}{2}\sum_{i=1}^{n}\left(\log\frac{\bar{\sigma}^2_{\theta_i}}{\sigma^2_\theta} + 1 - \frac{\bar{\sigma}^2_{\theta_i}}{\sigma^2_\theta} - \frac{(\bar{\mu}_{\theta_i} - \mu_\theta)^2}{\sigma^2_\theta}\right) \\
&\quad + \frac{1}{2}\sum_{j=1}^{d}\left(\log\frac{\bar{\sigma}^2_{\alpha_j}}{\sigma^2_\alpha} + 1 - \frac{\bar{\sigma}^2_{\alpha_j}}{\sigma^2_\alpha} - \frac{(\bar{\mu}_{\alpha_j} - \mu_\alpha)^2}{\sigma^2_\alpha}\right) \\
&\quad + \frac{1}{2}\sum_{j=1}^{d}\left(\log\frac{\bar{\sigma}^2_{\beta_j}}{\sigma^2_\beta} + 1 - \frac{\bar{\sigma}^2_{\beta_j}}{\sigma^2_\beta} - \frac{(\bar{\mu}_{\beta_j} - \mu_\beta)^2}{\sigma^2_\beta}\right) \\
&\quad + \frac{1}{2}\sum_{(i,j)\in\mathcal{I}}(2\mathbb{E}_q[z_{ij}] - \mu_{ij} - \bar{\mu}_{ij})\,(\mu_{ij} - \bar{\mu}_{ij}).
\end{aligned}$$

The last equality is up to constants that do not depend on the variational parameters ν.

Data and setup

Votes $\boldsymbol{y}$ from $n = 103$ senators across $d = 657$ bills are recorded, with $|\mathcal{I}| = 63254$ actual votes cast. We set 10% of these votes aside for model validation. Each $y_{ij} \in \{0, 1\}$ for $(i, j) \in \mathcal{I}$ indicates a vote "nay" or "yea" respectively by senator i on bill j.

When fitting our model, we choose prior hyperparameters $\mu_\theta = \mu_\beta = 0$, $\mu_\alpha = 1$, $\sigma^2_\theta = \sigma^2_\beta = \sigma^2_\alpha = 1$, with the understanding that the sheer amount of data will make a precise tuning of these weakly informative priors largely redundant. The prior on $\boldsymbol{\alpha}$ largely determines the scale of the ideological axis, and is somewhat arbitrary.

Results

We summarize the variational posteriors $q(\theta_i; \bar{\mu}_{\theta_i}, \bar{\sigma}^2_{\theta_i})$ of the senator ideologies in Figure. 1 (left). The means $\bar{\mu}_{\theta_i}$ are used to rank senators in order of left/right political inclination; the observed split aligns cleanly with party affiliations. The ideological positions of the 2 Independent senators seem to lean largely towards those of the Democratic party. As expected, the uncertainty in the posterior of θ_i is generally inversely related to the number of votes cast by senator i: see Cowan, Chiesa.

The bill landscape has been summarized in Figure 2, with the five most polarizing bills marked. Their ideological positions β_j seem to fall right on the line separating the Republican and Democratic senators. The main bulk of bills seem to fall to the left of this threshold and with positive α_j, indicating that it is likely that almost all Democratic senators (with larger ideological position β_i) would have viewed them favorably.

Diagnostics

The ELBO stabilizes within ≈ 100 iterations after a sharp monotonic increase (Figure 1, right), which gives some measure of confidence that CAVI updates have found a good max-

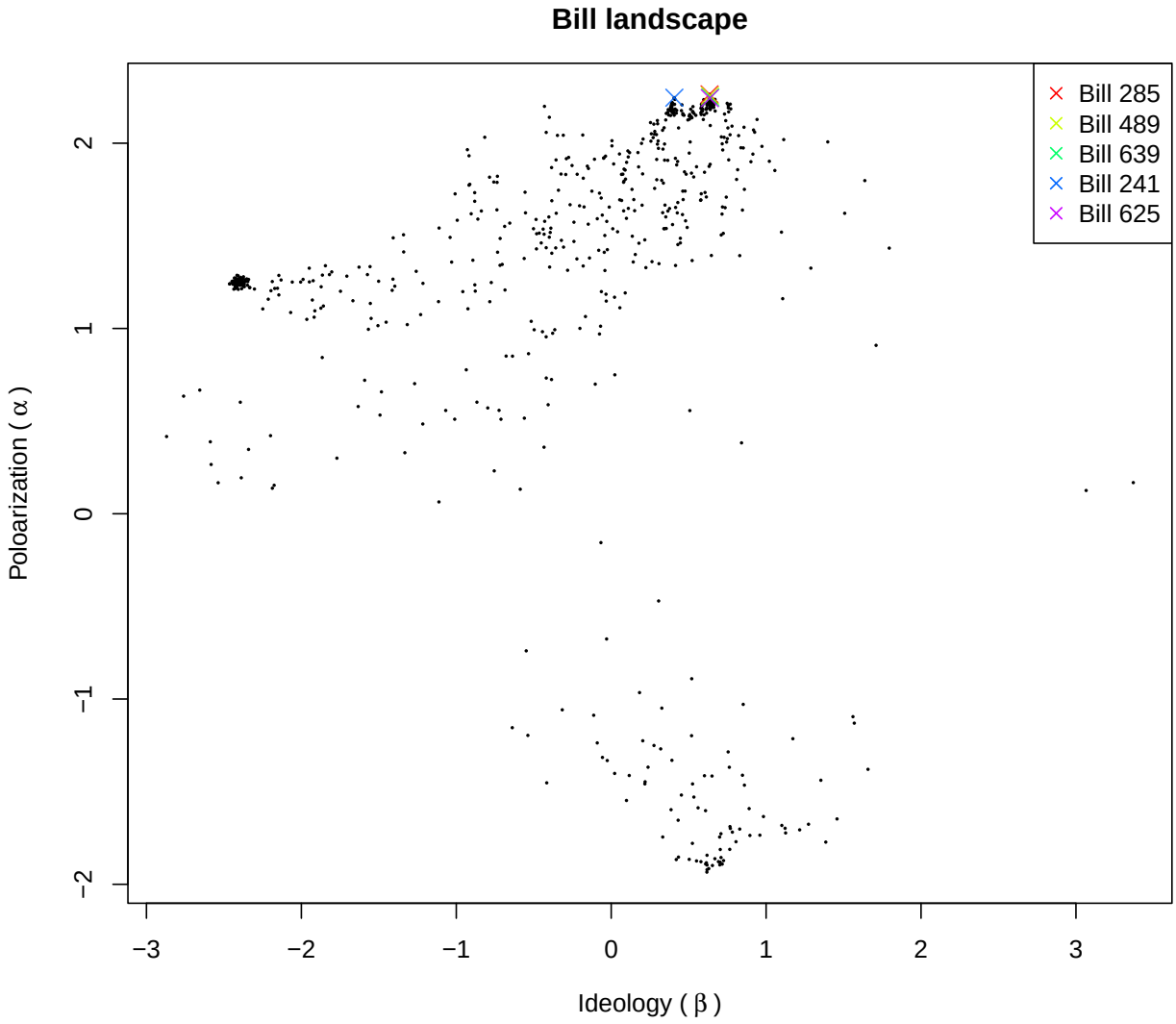


Figure 2: Approximate posterior ideological thresholds and polarizing powers of bills, depicted as $(\bar{\mu}_{\beta_j}, \bar{\mu}_{\alpha_j})$. The five most polarizing bills (with largest estimated magnitude of α_j) have been annotated.

imum. Qualitatively, the fitted model produces sensible interpretable results.

Evaluating the fitted model on the held-out set, $\{y_{ij}\}_{(i,j)\in\mathcal{I}_o}$ yields a prediction accuracy of 94.4%, and an average log posterior predictive density score of -0.144. A prediction for y_{ij} is obtained by rounding the posterior predictive probability

$$\mathbb{P}(y_{ij} = 1 \mid \boldsymbol{y}_{\text{in}}) = \int \Phi(\alpha_j(\theta_i - \beta_j))\, p(\theta_i, \alpha_j, \beta_j \mid \boldsymbol{y}_{in})\, d\theta_i\, d\alpha_j\, d\beta_j,$$

which in turn is approximated as a Monte-Carlo average over samples of $(\theta_i, \alpha_j, \beta_j)$ from q_{ν^*}. The posterior variances seem small enough that plugging in the point estimates $(\bar{\mu}_{\theta_i}, \bar{\mu}_{\alpha_j}, \bar{\mu}_{\beta_j})$ ought to give a decent estimate, $\Phi(\bar{\mu}_{\alpha_j}(\bar{\mu}_{\theta_i} - \bar{\mu}_{\beta_j}))$.

Question 4

Image segmentation generally describes the process of separating an image into different groups or regions of pixels, with pixels in a group sharing similar characteristics.. A crude method of image segmentation involves clustering pixels by their color, with no regard to their spatial arrangement. In this work, we explore simple ways of introducing spatial structure to a mixture model for segmenting color images; this is accomplished by imposing suitable priors on group labels. In particular, we focus on hierarchical Potts-Markov-like priors which encourage smoothing based on neighborhoods and group-specific temperatures. We further derive corresponding mean-field variational inference procedures and apply this method on the Berkeley Segmentation Dataset.